Electrical and Electronic Engineering, School of Engineering

EEE2008: Final Report

At the end of the project, you must prepare and submit a group final report / thesis on Canvas. In this report, you will discuss the detailed functional and theoretical analysis that was carried out into the individual subsystems you have created, and the overall integrated system. This report will be marked as a group project, and it is worth 40% of your final mark. However, this group mark will then be scaled depending on the outcome of the peer assessment to give the individual mark for the final report.
Note: to help the examiners, at the beginning of each section of the report please state who has worked on this section.

The report is limited to 4000 words (not including references / appendices), and it is suggested it is divided into the sections shown below. Please use a sensible font size, minimum 12 pt. An important consideration for writing a good report is to think about how much to write in each section based on the marks allocated. This does not have to be done arithmetically but more logically. Also, sometimes a lot of information can be easily presented in terms of a diagram, so if appropriate, use a diagram. However, when you insert an image, make sure you also reference the image somewhere in your text to discuss it, e.g., Looking at Fig. 1(a) etc…. Please sure they are also legible and informative, for example, not having the right legends on a figure can make it pointless.

Section	**Maximum Mark**
Abstract & Introduction	10
Theoretical Background, System Description, Results & Discussion of Subsystems	70
Conclusion	10
Presentation Quality / Referencing	10
Total	100

Suggested Sections

1) Abstract / Aims and Objectives

The abstract should be a paragraph summarising your main findings, methods of working, results, and conclusions. Also, write your overall project aims here, and show how you will achieve these aims through specific technical objectives, such as:
1) Objective 1,
2) Objective 2; etc… Use your own interpretation here.

2) Introduction / Overall System model

Describe the project and the autonomous line following buggy here. This is a good place to describe the overall system model with interactions and interfaces between different subsystems (without being too detailed). You can use the later subsystem sections to provide more detail.

3) Sensor Subsystem Theory & Implementation

Describe the related theory and design of the subsystem to a sensible detail, do not insert all of the information available in textbooks as where appropriate you can use references. The related practical / simulated implementations and results should also be presented with a thorough discussion. These results need to illustrate the performance of the subsystem. Measurements, scope traces etc... are appropriate, please do not just state that "it worked well". To achieve high marks, comprehensive and fully justified critical analysis is expected. Good cross-referencing between sections will makes a report look highly integrated and coherent. Also, by doing this you are avoiding unnecessary repetition.

4) Motor Subsystem Theory & Implementation

See comments for Sensor Subsystem above.

5) Control Subsystem Theory & Implementation

See comments for Sensor Subsystem above.

6) More Advanced Systems Theory & Implementation

Place any related theory and results for the extra subsystems that you have created such as the battery power system and the Bluetooth communications here.

7) Conclusion

A conclusion should summarise the work carried out, and the results and findings related to the original project objectives. Your observations and difficulties can be mentioned here, and you can discuss if your objectives were met.

8) References

Show any references you have used here. Please use the IEEE standard referencing style. i.e.:
[1] H. Kopka and P. W. Daly, A Guide to LATEX, 3rd ed. Harlow, England: Addison-Wesley, 1999.

EEE2008: Final Report Mark Sheet

Student Names:

Group:

Supervisor / Marker:

Please mark this as a group report.
Any students not contributing to this report without good reason will receive zero marks.

Section	Maximum Mark	Mark Obtained
Abstract & Introduction	10	
Theoretical Background, System Description, Results & Discussion of the Subsystems	70	
Conclusion	10	
Presentation Quality / Referencing	10	
Total	100	

Comments and Feedback

Abstract & Introduction

Excellent (9-10)	Good (6-8)	Developing (4-5)	Absent / Poor (0-3)	Mark
• An excellent and concisely written abstract, summarising the work undertaken for the project and the main findings. Includes all of the necessary information and is very professionally written. • The projects aims and objectives are clearly shown, identified, and justified. • An excellent introduction with extensive background information introducing the project, overall system, interfaces, etc… Evidence of significant relevant extra reading is shown.	• A very well written abstract, summarising the project and findings. Includes of all the necessary information and is professionally written. • The projects aims and objectives are clearly shown and discussed. • A good introduction was written which introduces the project. Information which is distinct from the initial project material is presented.	• A satisfactory to good abstract, written correctly and summarising the main target of the project, but has perhaps missed out a few things. • The projects aims and objectives and how they will be met are briefly discussed but more information is needed. • The introduction satisfactorily introduces the project, but more information needed to have been shown.	• An inappropriate abstract, which is either non-existent, incorrect, or too short to summarise the work undertaken, the results, and the main findings, adequately and accurately. • The projects aims and objectives and how they will be met are not discussed. • In the introduction the project is not introduced, or if it is, limited / incorrect information is given covering the overall system, interfaces, etc…	/10

Theoretical Background, System Description, Results & Discussion of the Subsystems
Please mark appropriately if excellent in one subsystem and poor in others.

Outstanding (56-70) (80%+)	Excellent (47-55) (67-79%)	Good (38-46) (54-66%)	Developing (28-37) (40-53%)	Absent / Poor (0-27) (0-39%)	Mark
• The marks for this section are obtained by presenting work that builds on that discussed in the 'Excellent' section. Extra research / reading is evident with work presented that improves the subsystems with novel design features. • Examples include Bluetooth communication, battery power monitoring, more complicated control methods other than PID, higher order filtering in the sensor subsystem, algorithms to reacquire the track, etc…	• An excellent and concise discussion into the three subsystems is shown. • All theory behind the subsystems is discussed in-depth showing a good knowledge of the fundamentals of the project areas. • An excellent set of test results / scope traces is presented for each subsystem. • These results are analysed and discussed in detail, with the responses compared back to the fundamental theory.	• A good discussion into the three subsystems is shown, although this is heavily based in the information given in the notes. • The theory behind the subsystems is discussed albeit with some errors. • Results show clear evidence of the methodical testing of the subsystems such as scope traces at various points in the circuits, plots of sensor responses with position, plots of control loop response with varying constants (tuning) and quantitative measures of car performance.	• A satisfactory explanation into the three subsystems is given, but a lot more information was needed. • No evidence of any more complicated work is shown. • Some theory and knowledge behind the different subsystems is presented, but there are errors in fundamental understanding. • Some results are presented but they do not show the full working of all of the subsystems.	• The explanation into the three main subsystems is poor / very limited or non-existent. • No evidence of any more complicated work is shown. • Limited theory / knowledge of the subsystems is presented, and there are fundamental mistakes in the understanding of them. • Very limited / no results are presented to explain the systems.	/70

Conclusion

Excellent (9-10)	Good (6-8)	Developing (4-5)	Absent / Poor (0-3)	Mark
• An excellent well written conclusion summarising all of the work undertaken, the methods, results, and the key technical findings. • Excellent discussion into the strengths / weaknesses of their designs, the performance of various circuits / sensors and recommended improvements to the design • Any particular research undertaken, and novel contributions added to the project are mentioned.	• A good conclusion, correctly written and summarising all of the project work undertaken and the main technical findings. • Some discussion into the strengths / weaknesses of their design, the performance of various circuits / sensors and recommended improvements to the design	• Relevant information covering some / most of the main work undertaken but has missed out minor points. • Limited discussion into the strengths / weaknesses. and performance.	• Conclusion is missing or is too short to adequately and accurately summarise the work undertaken. • Information presented is completely incorrect and inappropriately presented.	/10

Presentation Quality / Referencing

Excellent (9-10)	Good (6-8)	Developing (4-5)	Absent / Poor (0-3)	Mark
• An excellent report, with no notable areas of weakness. • Publications are relevant and proper, and referenced using IET/IEEE formatting. All references are cited in the text. • Report is very well written has no notable formatting errors. • Appropriately formatted figures, tables, captions, and equations (including numbering). • Excellent standard of written English throughout report.	• A good report. • Publications are correctly referenced using IET / IEEE formatting and cited in the text. • Report may have some minor formatting errors but is well written. • Appropriately formatted figures, tables, captions, and equations (including numbering). • Very good standard of written English throughout report. Only minor grammatical issues.	• A satisfactory report. • Publications are generally referenced and cited to an acceptable standard, but there may be some formatting issues, and minor referencing errors. • Some minor formatting errors with figures, tables, captions, and equation numbering. • Satisfactory standard of written English. Grammatical issues throughout, but they do not compromise the understanding of the report content.	• General presentation of report is very poor, and it is poorly organised throughout. • Very poor or non-existent reference list / Inadequate number of references. • Publications are poorly referenced, and not using IET / IEEE formatting. • Very poor figure quality. Missing axes titles /units / numbers • Figure and table captions are omitted throughout the report. • Equations are not numbered throughout the report. • Unacceptable standard of written English, which completely compromises the understanding of the report.	/10